

Position-Noise Models Used in `exp3/exp4`

Common notation

Let ψ denote the aircraft heading (`trk_rad`), g the groundspeed, and $\Phi(\cdot)$, $\varphi(\cdot; \sigma)$ the standard normal CDF and the zero-mean normal PDF with standard deviation σ . Along-track error e_a and cross-track error e_c rotate into East/North components via

$$e_E = e_a \sin \psi + e_c \cos \psi, \quad e_N = e_a \cos \psi - e_c \sin \psi. \quad (1)$$

All six models are calibrated so that the radial error $R = \sqrt{e_E^2 + e_N^2}$ satisfies $P(R \leq \text{ci95}) = 0.95$ for a given target ci95 (10 m or 3 m in the `exp3/exp4` sweep).

1 Normal (isotropic Gaussian)

$$(e_E, e_N) \sim \mathcal{N}(0, \sigma^2 I_2), \quad \sigma = \frac{\text{ci95}}{2.448}. \quad (2)$$

The constant $2.448 = \sqrt{-2 \ln 0.05}$ makes R Rayleigh-distributed with CDF $1 - e^{-r^2/2\sigma^2}$, which hits exactly 0.95 at $r = \text{ci95}$ by construction. At `ci95` = 10 m: $\sigma \approx 4.085$ m.

2 Latency (positional bias)

Same isotropic Gaussian, plus a deterministic along-track lag applied before rotation:

$$e_a \sim \mathcal{N}(-\ell g, \sigma^2), \quad e_c \sim \mathcal{N}(0, \sigma^2), \quad \ell = \text{LATENCY_S} = 0.1 \text{ s}, \quad (3)$$

i.e. $(e_E, e_N) \sim \mathcal{N}(-\ell g (\sin \psi, \cos \psi), \sigma^2 I_2)$ — the same σ as the normal model, now off-centre by ℓg along the (negative) reported track direction.

3 Heavy-tail (Gaussian mixture)

$$(e_E, e_N) \sim (1 - w) \mathcal{N}(0, \sigma_1^2 I_2) + w \mathcal{N}(0, \sigma_2^2 I_2), \quad \sigma_2 = k \sigma_1, \quad (4)$$

with $w = \text{TAIL_WEIGHT} = 0.10$, $k = \text{TAIL_RATIO} = 3.0$. σ_1 is solved by bisection from the mixture's Rayleigh survival function:

$$(1 - w) e^{-\text{ci95}^2/2\sigma_1^2} + w e^{-\text{ci95}^2/2\sigma_2^2} = 0.05. \quad (5)$$

At `ci95` = 10 m: $\sigma_1 \approx 2.776$ m, $\sigma_2 \approx 8.328$ m.

4 Anisotropic Gaussian

$$e_a \sim \mathcal{N}(0, \sigma_a^2), \quad e_c \sim \mathcal{N}(0, \sigma_c^2) \quad (\text{independent}), \quad \sigma_a = \rho \sigma_c, \quad \rho = \sqrt{\text{ANISO_VAR_RATIO}} = 3. \quad (6)$$

There is no closed form for the radial CDF once $\sigma_a \neq \sigma_c$, so σ_c is solved numerically from

$$\int_{-\text{ci95}}^{\text{ci95}} \varphi(x; \sigma_a) \left[2 \Phi \left(\frac{\sqrt{\text{ci95}^2 - x^2}}{\sigma_c} \right) - 1 \right] dx = 0.95. \quad (7)$$

At ci95 = 10 m: $\sigma_c \approx 1.675$ m, $\sigma_a \approx 5.024$ m (stdev ratio exactly 3, as required).

5 Latency + Anisotropic

Model 4's (e_a, e_c), with the same bias as Model 2 added to the along-track component before rotation:

$$e_a \sim \mathcal{N}(-\ell g, \sigma_a^2), \quad e_c \sim \mathcal{N}(0, \sigma_c^2), \quad (8)$$

with the same $\sigma_a \approx 5.024$ m, $\sigma_c \approx 1.675$ m as Model 4.

6 Heavy-tail + Anisotropic

A two-component mixture where *both* components share the along/cross ratio $\rho = 3$, and the tail component's axes are both scaled by $k = 3$ relative to the core component:

$$(e_a, e_c) \sim (1 - w) \mathcal{N}(0, \text{diag}(\sigma_{a1}^2, \sigma_{c1}^2)) + w \mathcal{N}(0, \text{diag}(\sigma_{a2}^2, \sigma_{c2}^2)), \quad (9)$$

$$\sigma_{a1} = \rho \sigma_{c1}, \quad \sigma_{c2} = k \sigma_{c1}, \quad \sigma_{a2} = \rho k \sigma_{c1}. \quad (10)$$

σ_{c1} is solved from the mixed version of Eq. (7):

$$(1 - w) \text{RadialCDF}(\text{ci95}; \rho \sigma_{c1}, \sigma_{c1}) + w \text{RadialCDF}(\text{ci95}; \rho k \sigma_{c1}, k \sigma_{c1}) = 0.95. \quad (11)$$

At ci95 = 10 m: $\sigma_{c1} \approx 1.259$ m, $\sigma_{a1} \approx 3.778$ m, $\sigma_{c2} \approx 3.778$ m, $\sigma_{a2} \approx 11.334$ m.

Solved parameter values

Source: implemented in `sim_models/noise_distributions.py` (functions `gaussian`, `make_mixture_gaussian`, `make_anisotropic_gaussian`, `make_anisotropic_mixture_gaussian`); constants defined in `experiments/config.py`.

Model	ci95 = 10 m	ci95 = 3 m
Normal / Latency: σ	4.085	1.225
Heavy-tail: σ_1	2.776	0.833
Heavy-tail: σ_2	8.328	2.498
Anisotropic: σ_c	1.675	0.502
Anisotropic: σ_a	5.024	1.507
Heavy-tail + Anisotropic: σ_{c1}	1.259	0.378
Heavy-tail + Anisotropic: σ_{a1}	3.778	1.133
Heavy-tail + Anisotropic: σ_{c2}	3.778	1.133
Heavy-tail + Anisotropic: σ_{a2}	11.334	3.400

Table 1: Solved standard deviations (metres) for both POS_CI95_LEVELS used in exp3/exp4. LATENCY_S = 0.1 s, TAIL_RATIO = 3, TAIL_WEIGHT = 0.10, ANISO_VAR_RATIO = 9 ($\rho = 3$) for both.